Vancomycert: A Certified Neuro-Symbolic Drug Delivery System

UK Robotics Summer School 2026

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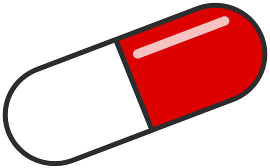
Overview

- Problem description
- Mathematical Formalisation
- Vehicle Specification
- Summary

Outline

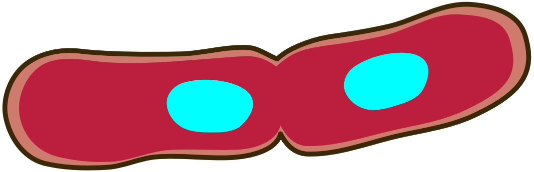
Vancomycin

- Vancomycin is an antibiotic



Vancomycin

- Vancomycin is an antibiotic
- Treats severe and resistant bacterial infections



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Vancomycin

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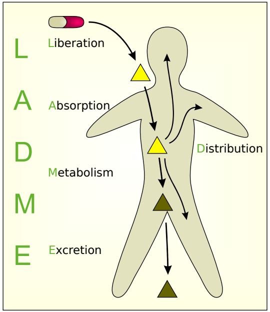
- Treats severe and resistant bacterial infections
- widely perscribed
- **31.7% of patients experienced nephrotoxicity[1]**

[1] Al-Maqbali et al. – "Vancomycin Therapeutic Drug Monitoring (TDM) and Its Association with Clinical Outcomes"

Pharmacokinetics and Pharmacodynamics Modelling

- Pharmacokinetics is what the body does to the drug



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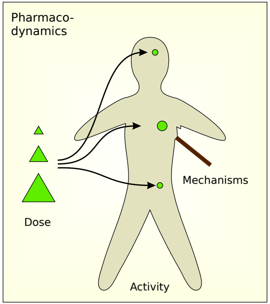

Pharmacokinetics and Pharmacodynamics Modelling

- Pharmacodynamics

Pharmacokinetics and Pharmacodynamics Modelling

- Pharmacodynamics $\rightarrow$ Optimisation

Pharmacokinetics and Pharmacodynamics Modelling

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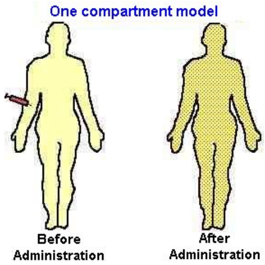
Pharmacokinetics and Pharmacodynamics Modelling

- Pharmacodynamics $\rightarrow$ Optimisation $\rightarrow$ Machine Learning
- Pharmacokinetics $\rightarrow$ Safety-Critical $\rightarrow$ Formal Methods

Compartment Models

- **One Compartment:** The body acts as one single system, the drug is absorbed by the entire body.



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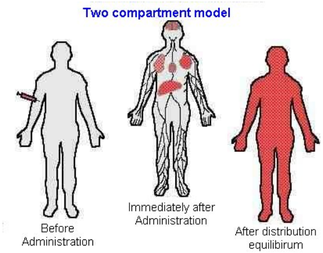
- **Two Compartment:** The organs first absorb the drug and deposit it in the blood, then the rest of the body processes it.



Compartment Models

- **One Compartment:** The body acts as one single system, the drug is absorbed by the entire body.
- **Two Compartment:** The organs first absorb the drug and deposit it in the blood, then the rest of the body processes it.
- We focus on the One Compartment model for simplicity

References

Al-Maqbali, Juhaina Salim et al. "Vancomycin Therapeutic Drug Monitoring (TDM) and Its Association with Clinical Outcomes: A Retrospective Cohort". In: Journal of Infection and Public Health 15.5 (May 1, 2022), pp. 589–593. ISSN: 1876-0341. DOI: 10.1016/j.jiph.2022.04.007. URL: https://www.sciencedirect.com/science/article/pii/S1876034122000910 (visited on 06/17/2026).

Outline

Pharmacokinetic Constants

Constants	Representation

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k_a	Absorption Coefficient

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Variables	Representation
C	Concentration of Vancomycin
C_Σ	Total Concentration
D	Dose size
$\bar{D}$	List of all doses
t	Time size first dose

One Compartment Equation

$$C(D,t) = \frac{D \cdot k_a}{Vd \cdot (k_a - k_e)} \cdot \left(e^{-k_e \cdot t} - e^{-k_a \cdot t}\right)$$

One-Compartment Pharacokinetic model for oral medication[2]

[2] Talevi and Bellera – "One-Compartment Pharmacokinetic Model"

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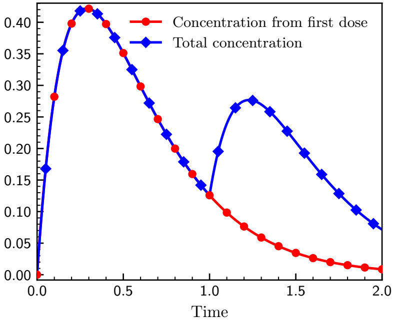
One-Compartment Pharacokinetic model for oral medication[2]

$$C_\Sigma(\bar{D},t) = \sum_{i=0}^{n} (\max(0, C(\bar{D}_i, t - ttd \cdot i)))$$

Total concentration[3]

[2]Talevi and Bellera – "One-Compartment Pharmacokinetic Model"
[3]Sirman et al. – Vancomycert: A Certified Neuro-Symbolic Drug Delivery System (Case Study)

One Compartment Equation



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C does not depend on any of these[4]

[4] C does depend on weight, but it is assumed to be constant and known statically

Neural Network Involvement

An initial dose is given, then n more doses with interval ttd

$$doses \colon \mathbb{R} \to \mathbb{N} \to list\ \mathbb{R}$$

$$doses(c, n) = \begin{cases} [\mathcal{N}(c)] & n = 0 \\ doses(c, n-1) + [:: \mathcal{N}(C_\Sigma(doses(c, n-1), ttd \cdot n))] & n > 0 \end{cases}$$

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$$\mathit{doses} \colon \mathbb{R} \to \mathbb{N} \to \mathit{list}\ \mathbb{R}$$

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Main Safety Definition

Given that the concentration is initially safe, then, for all time, for any number of doses, the total concentration will never exceed the safe boundry.

$$\forall c : \mathbb{R}, c \leq C_{safe} \implies \forall t : \mathbb{R}, \forall n : \mathbb{N}, C_{\Sigma}(doses(c, n), t) \leq C_{safe}$$

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Cannot naively be used in vehicle

Property for Vehicle

The current concentration is used as an input to the neural network, along with the time at which concentration peaks, can be used to calculate the maximum change (d), which can be added to the current concentration.

$$\forall c : \mathbb{R}, c \leq C_{safe} \implies c + C\left(\mathcal{N}(c), \frac{\ln\left(\frac{k_a}{k_e}\right)}{k_a - k_e}\right) \leq C_{safe}$$

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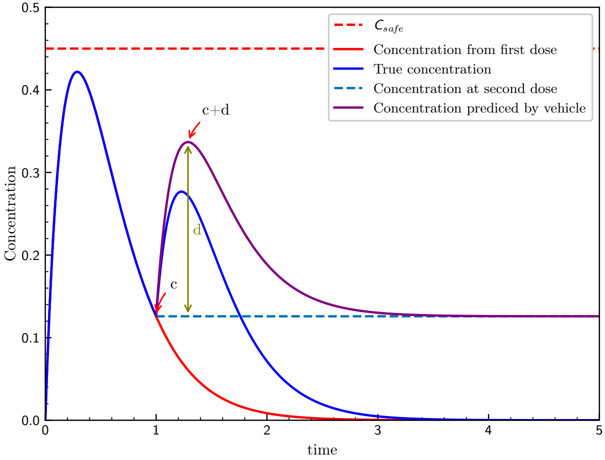

Not Quite...

$$\forall c : \mathbb{R}, c \leq C_{safe} \implies c + C\left(\mathcal{N}(c), \underbrace{\frac{\ln\left(\frac{k_a}{k_e}\right)}{k_a - k_e}}_{\text{Irrational!}}\right) \leq C_{safe}$$

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Approximate!

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Solvers can't handle irrational numbers...
Approximate!

$$\text{Ke\_under} \leq e^{-k_e \cdot \frac{\ln\left(\frac{k_a}{k_e}\right)}{k_a - k_e}} \leq \text{Ke\_over}$$

$$\text{Ka\_under} \leq e^{-k_a \cdot \frac{\ln\left(\frac{k_a}{k_e}\right)}{k_a - k_e}} \leq \text{Ka\_over}$$

Not Quite...

If $k_a < k_e$:

$$e^{-k_e \cdot \frac{\ln\left(\frac{k_a}{k_e}\right)}{k_a - k_e}} - e^{-k_a \cdot \frac{\ln\left(\frac{k_a}{k_e}\right)}{k_a - k_e}} \geq \text{Ke\_under} - \text{Ka\_over}$$

else:

$$e^{-k_e \cdot \frac{\ln\left(\frac{k_a}{k_e}\right)}{k_a - k_e}} - e^{-k_a \cdot \frac{\ln\left(\frac{k_a}{k_e}\right)}{k_a - k_e}} \leq \text{Ke\_over} - \text{Ka\_under}$$

The change in sign is to respect the effect of the fraction in C.

$$C(D, t) = \frac{D \cdot k_a}{Vd \cdot (k_a - k_e)} \cdot \left(e^{-k_e \cdot t} - e^{-k_a \cdot t}\right)$$

References

- Sirman, Alistair et al.
  Vancomycert: A Certified Neuro-Symbolic Drug Delivery System (Case Study).
  2026. arXiv: 2606.19532 [cs.LO]. URL: https://arxiv.org/abs/2606.19532.
- Talevi, Alan and Carolina L. Bellera. "One-Compartment Pharmacokinetic Model". In:
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  Cham: Springer International Publishing, 2021, pp. 1–8. ISBN: 978-3-030-51519-5.
  DOI: 10.1007/978-3-030-51519-5_58-1. URL:
  https://doi.org/10.1007/978-3-030-51519-5_58-1.

Outline

Preamble


```
type UnnormalisedInputVector = Tensor Real [5]
type InputVector = Tensor Real [5]

conc = 0
temp = 1
wbc = 2
age = 3
weight = 4

type OutputVector = Tensor Real [1]
```

Normalisation


```
@dataset
meanScalingValues : UnnormalisedInputVector

@dataset
standardDeviationValues : UnnormalisedInputVector

normalise : UnnormalisedInputVector -> InputVector
normalise x = foreach i .
  (x ! i - meanScalingValues ! i) / (standardDeviationValues ! i)
```

Network


```
@network
pk : InputVector -> OutputVector

normpk : UnnormalisedInputVector -> OutputVector
normpk x = pk (normalise x)
```

Parameters


```
@parameter
Ka, Ke, Vd, C_safe, ttd,
  Ka_over, Ka_under,
  Ke_over, Ka_under : Real⁴

@property
Ka_pos : Bool
Ka_pos = 0 < Ka

@property
Ke_pos : Bool
Ke_pos = 0 < Ke

@property
Ke_n_Ka : Bool
Ke_n_Ka = Ka != Ke

@property
Vd_pos : Bool
Vd_pos = 0 < Vd

@property
C_safe_pos : Bool
C_safe_pos = 0 < C_safe
```

[4] This is not valid syntax, but it is shortened for ease of reading.

Precondition


```
safeInput : Input -> Bool
safeInput x =
  0    <= x ! conc   <= C_safe and
  36.5 <= x ! temp   <= 40     and
  7.5  <= x ! wbc    <= 20     and
  18   <= x ! age    <= 89     and
  50   <= x ! weight <= 100
```

Main Safety Property


```
safeOutput : Input -> Bool
safeOutput x =
  let y = ((((normpk x) ! 0) * Ka) / (Vd * (Ka - Ke)))
  in if Ka < Ke
   then (x ! conc) + y * (Ke_under - Ka_over) <= C_safe
   else (x ! conc) + y * (Ke_over - Ka_under) <= C_safe

@property
safe : Bool
safe = forall x . safeInput x => safeOutput x
```

Non Negativity


```
nonNegOutput : Input -> Bool
nonNegOutput x = 0 <= (normpk x) ! 0

@property
nonNeg : Bool
nonNeg = forall x . safeInput x => nonNegOutput x
```

Outline

Vehicle for more than just robustness

- Parametric over patients

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- Parametric over patients
- Normalisation

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- Integration with proof assistants

Vehicle for more than just robustness

- Parametric over patients
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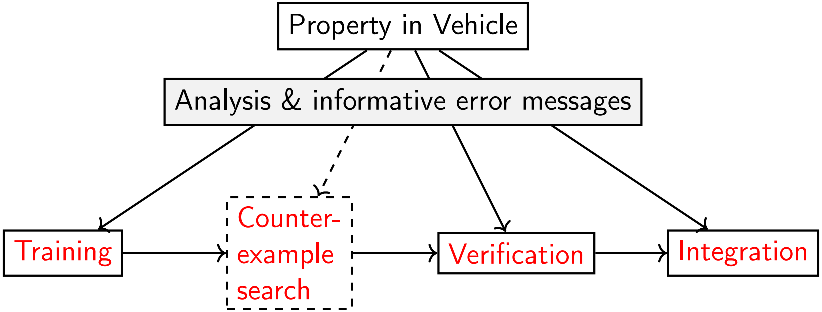
- **Property Driven Training**

Vehicle for more than just robustness



Training	Counter-example search	Verification	Integration
DL2 or other DL (native)	PGD FGSM etc.	Marabou Eran etc.	Lean Rocq Agda etc.

Why do all this?

- Ease of use

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- Formalises requirements

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- Standardisation

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- Ease of use
- Formalises requirements
- Standardisation
- Mathematical proofs

Vancomycert vs AcasXu

- 1-step vs Infinite-horizon property

Vancomycert vs AcasXu

- 1-step vs Infinite-horizon property
- Many networks working in tandem vs one at a time

Vancomycert vs AcasXu

- 1-step vs Infinite-horizon property
- Many networks working in tandem vs one at a time
- Both safety critical

Find the paper

Sirman et al. – Vancomycert: A Certified Neuro-Symbolic Drug Delivery System (Case Study)

Next Up...

How to train a network that meets the specification